

Week 7 Lesson Plan

Enterprise RAG: The Measure of All Things — Evaluating RAG Systems from Retrieval to Reasoning

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The Measure of All Things: Evaluating RAG Systems from Retrieval to Reasoning

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When you can measure what
you are speaking about and
express it in numbers, you
know something about it; but
when you cannot measure it,
your knowledge is of a meagre
and unsatisfactory kind.

Lord Kelvin, 1883

Not everything that counts can
be counted, and not everything
that can be counted counts.

William Bruce Cameron, 1963

A Gentle Re-Entry

If you need a refresher on the two gates, re-visit last week’s lesson plan: *The Two Gates of the Gatehouse*.

LAST WEEK WE HUNG TWO GATES ON THE LIBRARY DOOR. The request gate—the gatehouse—learned to keep bad questions out: the four threat families, deobfuscate-then-inspect, ACL bound into retrieval, the cheap-first cascade. The response gate—the conscience—learned to keep bad answers in: the assertion–evidence bipartite graph, the escalation ladder, the trichotomy of answer, hedge, and refuse. We closed on epistemic humility: the graded partial answer, the honest “I don’t know.”

Both gates were *architectural*: we engineered the *mechanism* for trustworthiness. But mechanism without measurement is hope. How do we know the guardrails actually fire when they should? How do we know the grounding pipeline catches unsupported claims rather than waving them through? How do we even know that the retrieval pipeline we spent five weeks refining is finding the right documents, in the right order?

This week, we stop building and start measuring. Measurement is not a final act—it is the foundation that every prior week has been resting on, unexamined. Today we make it explicit.

ONE THREAD DESERVES SPECIAL ATTENTION. Last week’s two gates gave the system the *ability* to abstain and the *ability* to verify. This week completes that arc: we build the *metrics* for epistemic humility—negative rejection rate, noise robustness, abstention calibration—and connect them to the architectural decisions we have already made. A system that knows when it does not know, and can *prove* that it knows when it does not know, is a system you can trust in production.

What Must You Carry Forward

WHEN TODAY’S SESSION CONCLUDES, you should be able to:

- Construct an EVAL dataset: select queries from SMEs, assign graded relevance scores, enforce scale (minimum 200, shoot for 1,000), and manage eval decay as the production corpus evolves.
- Explain why evaluation is the AI engineer’s own concern—not QA’s, not a downstream team’s—and why AI systems admit no clean pass/fail.
- Compute the six classical retrieval metrics—Recall@K, Precision@K, MRR, MAP, AUC-PR, and nDCG—plus F_1 , and explain the progression from binary to rank-aware to relevance-weighted evaluation.
- State the optimisation goal $\arg \max_{\theta} \mathbb{E}_{q \sim \mathcal{Q}} [\mu(q, \theta)]$ and explain what θ , μ , and $\mathcal{Q}$ represent in a RAG system.
- Decide whether a metric improvement is real: run a paired bootstrap or permutation test, and distinguish statistical from practical significance.
- Describe the four core RAGAS metrics—faithfulness, answer relevancy, context precision, context recall—and explain the limitations of each, including circular LLM dependency and lack of claim-level granularity.
- Decompose a generated answer into atomic claims and evaluate each via FActScore-style claim-level entailment; explain what ALCE adds on citation quality.

WHY? Six weeks of geometry, retrieval, chunking, artifacts, graphs, and gates culminate in a single principle: *you cannot improve what you cannot measure*. The metrics you choose determine what your system optimises for; the eval set you build determines whether your measurements mean anything; the evaluation framework you adopt determines whether you can diagnose failures and fix them. Get evaluation wrong, and every prior week’s work is ungrounded. Get it right, and the entire pipeline becomes a system you can steer.

- Explain the LLM-as-a-Judge paradigm: direct assessment vs. pairwise ranking, the role of Prometheus-2 and its kin, the judge biases (position, length, self-enhancement, style-over-substance), and how to calibrate a judge against human labels with Cohen's κ .
- Describe the four RGB abilities—noise robustness, negative rejection, information integration, counterfactual robustness—and explain why they matter for production RAG systems.
- Compute expected calibration error (ECE), read a reliability diagram, and name the standard calibration fixes.
- Design an evaluation framework for agentic RAG that goes beyond final-output accuracy to include trajectory quality, tool-use accuracy, and planning trace efficiency.
- State Goodhart's law, recognise the RAG-specific reward-hacking modes, and apply the standard Goodhart-proofing techniques.
- Connect the evaluation dimensions back to every prior week of this summer: retrieval quality to Weeks 1–2, chunking to Week 3, derivative artifacts to Week 4, graph reasoning to Week 5, and the two gates to negative rejection and faithfulness.

If you can do these without hand-waving, you have crossed from RAG practitioner to RAG engineer.

Act I: The Yardstick—Before Any Metric, a Ground Truth

YOU CANNOT MEASURE WITHOUT A YARDSTICK. When we take a ruler off the desk, the little notches are the metric; but the ruler itself is the yardstick, the physical artefact against which observations are compared. Without it, every statement about length collapses into opinion: *longer than*, *shorter than*, *pretty short*. In information retrieval, the yardstick is a **gold dataset**, and before we talk about metrics we must talk about how to build it—and whose job that building really is.

The EVAL Set: Foundation of All Measurement

BEFORE ANY METRIC CAN BE COMPUTED, YOU NEED GROUND TRUTH. An EVAL set is a curated dataset of queries $\mathcal{Q} = \{q_1, q_2, \dots, q_n\}$, where each query q_i is annotated with a set of relevant documents $\{d_{i1}, d_{i2}, \dots\}$ and associated relevance scores $\{r_{i1}, r_{i2}, \dots\}$. Relevance scores are typically discrete: $r \in \{0, 1, 2, 3, 4\}$, where 0 is irrelevant and 4 is perfectly relevant.

THERE ARE TWO WAYS TO BUILD IT, and they differ in cost by orders of magnitude. The first is the *gold standard*, as practised by Google and Microsoft: throw a large query catalogue at the engine, and have teams of human evaluators tick-mark every returned result; for recall, list what *should* have been returned and compare. Watertight—and unaffordable for a typical enterprise pod of seven engineers.

The second way is the **cherry-picked gold dataset**, and it works. The recipe: (1) hire subject-matter experts, not engineers—clinicians for healthcare, professors for learning management, compliance officers for finance; (2) have the SMEs author queries harvested from real intent (search logs, support tickets); (3) throw a simple engine at each query and keep the top 100–200 candidates—a tractable scan, unlike a million documents; (4) have the SMEs cherry-pick and *grade* the best 25–30 on the 0–4 scale; (5) freeze the dataset at each release, version it, and evolve it.

HOW LARGE MUST IT BE? Minimum 200 queries; shoot for 1,000. Below 200, patchiness dominates—you will crown one system over another on noise alone. At 1,000, a 1–2 point difference in nDCG is usually statistically significant. The metric you compute is an *estimate* of the expectation

$$\mathbb{E}_{q \sim \mathcal{Q}}[\mu(q)] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \mu(q_i),$$

and a small n gives a noisy estimate. (The Interlude will show that this rule of thumb and formal power analysis agree almost exactly.)

TWO FAILURE MODES THREATEN YOUR EVAL SET. First, **eval decay**: as the production corpus evolves, the relevance judgements go stale; a document supremely relevant in January is superseded in June. Refresh on a schedule. Second, **overfitting to the EVAL set**: if engineers know the queries, even implicitly, they will optimise for those queries. Keep a held-out test set that only the evaluation lead controls.

The gold standard has its own pathology. Plant ten beautiful articles on the Maxwell–Boltzmann distribution as ground truth; the engine surfaces an even better article you never catalogued; precision drops. The engine is penalised for doing its job. The moment humans enter the loop, nuance enters with them.

The “SME Summit” is the seed: gather 5–10 domain experts, provide breakfast and lunch, give them 8 hours, and have them produce 100 well-debated queries with graded relevance judgements. Grow it over the next six months.

A logicians' joke. Three people on a train see a white sheep in a field. One: "the sheep in this country are white." The second: "all you can say is that *at least some* sheep here are white." The third: "all you can say is that there exists at least one sheep, on at least one side of its body, that appears white at the moment of observation." Public benchmarks license only the third kind of claim.

WHY? A bridge engineer does not outsource load-testing, because the load curves *are* the design. The same is true of RAG: the metrics you choose shape the system you build; the eval set defines what the system optimises for; the thresholds decide what ships. Evaluation is not downstream of AI engineering. *Evaluation is AI engineering.* Skip it, and you have not built a production system; you have built a demo.

The corollary from every serious AI project: 80% of the effort goes to clean data, demonstrations, and the yardstick; 20% to the fashionable part—architecture, algorithm, model. At Cornerstone (100M learning objects, 7,000 clients), forty SMEs at two days each produced a 10,000-query gold dataset—and when the hybrid-search product launched, one big client was *furious* that search finally worked. There is no better sign that your evaluation methodology is real than an angry client who finally sees the improvement.

Against the Borrowing of Public Benchmarks

A COMMON TEMPTATION IS TO SKIP THE YARDSTICK WORK ENTIRELY by evaluating on a public benchmark. "MS MARCO is right there." This is like measuring how well a medicine works by testing it on fever patients, when your patients do not have fever. Your RAG may perform brilliantly on the benchmark and catastrophically on your users. Public benchmarks are good for one thing: a sanity check that your retriever is baseline-competent. They are not evidence that it will work in production.

Whose Job Is Evaluation?

A PATTERN WATCHED FOR TWENTY YEARS DESERVES NAMING. Ask a room of software engineers what they associate with *evaluation*, and the honest answer is usually: *QA's job*. Testing is what the test team does after the feature is written; evaluation is a handoff, downstream of the real work. This mental model is sticky—teams of senior engineers nod through an entire lecture on retrieval metrics, build enterprise RAG systems for months, and then confess, with genuine surprise, that they never implemented a gold dataset. The formulas did not leak out of their heads because they are lazy. They leaked because they were filed, in the mental filing cabinet, under *someone else's concern*. And that cabinet leaks.

IT IS THE WRONG MODEL FOR AI SYSTEMS. Conventional software is deterministic: same input, same output; QA writes assertions and CI runs them red or green. An AI system returns a *distribution of answer qualities* against a *distribution of queries*—and both distributions drift under you: model versions, prompts, stochastic decoding, corpus growth. There is no clean pass/fail to hand off.

The Optimisation Goal

ALL RAG TUNING REDUCES TO A SINGLE EQUATION. Let θ represent the full configuration of your RAG pipeline—chunk size, embedding model, retrieval strategy, reranker weights, prompt template, generation model, temperature, and every other knob. Let $\mu(q, \theta)$ be a metric of goodness for query q under configuration θ . The goal:

$$\arg \max_{\theta} \mathbb{E}_{q \sim Q} [\mu(q, \theta)]$$

We want the configuration that maximises expected performance across all queries in our EVAL distribution. This framing treats the RAG system as a black box and adjusts θ to improve μ .

Act II: The Staircase—The Six Classical Retrieval Metrics

THE SECOND ACT ESTABLISHES THE MATHEMATICAL VOCABULARY OF RETRIEVAL EVALUATION. For six weeks we have retrieved documents, passages, artifacts, and graph neighbourhoods. But we have never rigorously asked: did we retrieve the *right* things? In the right order? With the right degree of relevance? Six metrics answer these questions, and they build on each other in a logical progression—from binary to rank-aware to relevance-weighted. To build intuition, we thread two analogies through the entire progression: one from a beach, one from a cave.

The Seashell Analogy. Imagine a parent and two children on a beach.¹ The parent (the *evaluator*) throws a bag of seashells onto the sand—some buried, some in plain sight, mixed with pebbles that look like shells if you squint. Each child (a *search engine*) gets a bucket: *go get me ten seashells*. Call the children Sarah and Meera. Sarah returns with 4 shells and 6 pebbles; Meera with 7 shells and 3 pebbles. Everybody agrees Meera did better—but *how much* better, numerically?

As a child keeps scooping, two things happen simultaneously. The number of good seashells found can only increase, so **recall keeps rising**. But the bucket also fills with debris, so **precision keeps falling**. This is the precision–recall tradeoff made physical.

Metric 1: Precision@K. Of the K results returned, what fraction are relevant?

$$\text{Precision@K} = \frac{|\{\text{relevant docs in top } K\}|}{K}$$

Sarah scored $4/10 = 0.40$; Meera $7/10 = 0.70$. Precision measures *purity*: how clean is the result set? The mnemonic: **Precision emphasises Positive results (P and P)**.

Precision is *cutoff-sensitive*: ask for twenty shells instead of ten and the children’s ranking could reverse. And it says nothing about what was left behind in the sand.

This is the same framing as hyperparameter tuning in machine learning. The difference is that θ in RAG is much richer: not just model parameters but architectural choices, chunking strategies, retrieval pipelines, and prompt designs. The search space is vast, which is why systematic evaluation matters.

¹ This analogy maps directly to the precision–recall tradeoff and will reappear for AUC-PR.

An apple farmer searching for optimal harvest conditions types “apple farming” and gets Apple MacBook reviews and AAPL stock analysis. Topically irrelevant results that a human finds infuriating—precision quantifies this annoyance factor.

Deep Dive: A subtle point about maximum

achievable precision: if a query has $n = 10$ relevant documents and you retrieve $K = 20$, the *maximum possible* Precision@20 is $10/20 = 0.50$, even for a perfect system. This is not a failure—it is a mathematical ceiling imposed by $n < K$. Precision@K is meaningful primarily when K is small enough that a perfect system could approach 1.0, or when comparing two systems at the same K . Always report n and K together.

Metric 2: Recall@K. Of the n documents known to be relevant, how many appear in the top K ?

$$\text{Recall@K} = \frac{|\{\text{relevant docs in top } K\}|}{n}$$

Bury five *blue* seashells and ask for the blue ones: find three, and recall is $3/5 = 0.60$. Recall measures *completeness*: did we find everything? The mnemonic: **Recall emphasises Relevance (R and R)**. Note the asymmetry with precision: the denominator is the whole of the relevant world—which only the gold dataset knows.

Two family analogies for recall. Rookie, the golden retriever, sent to fetch a ball and returning with a chew toy: of the five balls scattered in the yard, the fraction of *balls* she returns is her recall. And the neuropsychologist daughter who plants facts early in a conversation and, two hours later, asks: “What did I tell you about my day?” Retrieving what was planted is recall—it is how we test the minds of ageing parents and the retrievers of search engines.

RECALL HAS A SUBTLE DEPENDENCE ON K . If ten things were planted but you may name only three, your recall caps at 0.30 however good your memory. At the crossover $K = n$, perfect precision and perfect recall coincide. And Recall@K is *monotonically non-decreasing* in K : one more scoop may add a shell or a pebble, but never removes a shell already in the bucket. This is why retrieval systems set K generously.

WHEN TO OPTIMISE RECALL FIRST. In legal RAG, missing the one precedent that wins the case is catastrophic—better to retrieve 50 noisy documents containing every precedent than 5 pristine ones missing the one that matters. In medical RAG, a missed drug interaction could harm a patient. Optimise recall first, clean up precision later—that is Week 2’s two-phase architecture: Phase 1 (retrieval) casts the wide net; Phase 2 (the reranker) pushes the seashells to the top of the bucket.

The pair, combined: F_1 . Neither precision nor recall alone tells the whole story: a system returning one always-correct document has $P = 1.0$ and abysmal recall; a system returning the entire corpus has $R = 1.0$ and no precision. We quote them together, or combine them in the harmonic mean:

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

The harmonic mean punishes imbalance: $P = 1.0, R = 0.1$ gives $F_1 \approx 0.18$; the balanced $P = R = 0.5$ gives $F_1 = 0.5$. You cannot buy a good F_1 with one perfect half.

Two limitations of the simple metrics. Precision, recall, and F_1 share two blind spots: (1) they are **not rank-aware**—three relevant documents at positions 1, 2, 3 score identically to the same three at positions 48, 49, 50; (2) they are **not degree-of-relevance-aware**—the relevant chapter of *Feynman’s Lectures* and a dry handbook entry count the same. The next four metrics fix these blind spots, one at a time.

The Cave of Ali Baba. To build intuition for rank-aware metrics, consider Ali Baba entering the cave of the forty thieves.² The loot is laid out linearly on shelves, establishing a **rank** for each item. But Ali Baba is in danger: the thieves may return. *Time is at a premium.* He needs to find one treasure and run. A fortunate day is when the first shelf holds a diamond; an unfortunate one is when he must walk past ninety-nine tools to find a silver coin. *How quickly you find the first treasure* maps directly to our next metric.

Metric 3: Mean Reciprocal Rank (MRR). For each query, find the rank of the *first* relevant result and take the reciprocal. Average across all queries:

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}$$

First shelf relevant: $1/1 = 1.0$. Third: $1/3 \approx 0.33$. Hundredth: 0.01 —the system is essentially not helping. A worked example: three queries whose first relevant results sit at ranks 1, 2, and 5 give $\text{MRR} = \frac{1}{3}(1 + \frac{1}{2} + \frac{1}{5}) \approx 0.57$.

WHEN TO USE MRR. Navigational queries—“What is the returns policy?”, “Where is the API reference?”—where the user wants *one* specific answer fast. But MRR has a critical limitation: it ignores everything after the first relevant result. If a query has 10 relevant documents, MRR cannot tell whether the system found 1 or all 10. Ali Baba grabbed the first bag and ran—he did not take inventory of the cave.

Metric 4: Mean Average Precision (MAP). Ali Baba had a brother—Qasim, greedy and impatient. When he enters the cave he wants *all* of it: every diamond, every ruby, as early in his walk as possible. He is the user doing research. MAP measures Qasim’s satisfaction: for a single query, compute the precision at every rank where a relevant document appears, then average those precision values:

$$\text{AP}(q) = \frac{1}{|\text{relevant}|} \sum_{k=1}^K P(k) \cdot \text{rel}(k) \quad \text{MAP} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \text{AP}(q_i)$$

where $P(k)$ is the precision at rank k and $\text{rel}(k)$ indicates relevance at rank k . Equivalently: at the i -th relevant document, found at rank r_i , the precision is i/r_i —the treasures so far over the shelves walked so far.

A WORKED EXAMPLE.³ Consider a query with 3 relevant documents in a top-5 list, at ranks 2, 4, and 5:

² The cave is the ranked result list; the shelves are ranks; jewels are relevant documents; tools and rope are irrelevant ones.

Ranking matters because users do not scan uniformly. As the old search-engineer joke goes: governments hide their national secrets on page two of the search results. Nobody ever visits.

The reciprocal is convex: the jump from rank 1 to 2 halves the score; from 10 to 11 it barely registers. All the metric’s sensitivity lives in the early ranks—where the user’s patience lives. $\text{MRR} > 0.3$ roughly means the first relevant result typically appears in the top 3.

³ The step from MRR to MAP is where most students stumble. Work this through by hand once, and it stays.

Rank	Relevant?	Precision@rank	Contribution to AP
1	No	—	0
2	Yes	$1/2 = 0.50$	0.50
3	No	—	0
4	Yes	$2/4 = 0.50$	0.50
5	Yes	$3/5 = 0.60$	0.60

$AP = \frac{1}{3}(0.50 + 0.50 + 0.60) = \frac{1.60}{3} \approx 0.533$. Note the double meaning of “mean average”: AP averages precision values *within* a query; MAP averages AP *across* queries.

Deep Dive: Position matters, quantitatively.

Two systems each find the same two relevant documents in a top-6 list. System A places them at ranks 1 and 4: $AP = \frac{1}{2}(1 + \frac{2}{4}) = 0.75$. System B places them at ranks 3 and 6: $AP = \frac{1}{2}(\frac{1}{3} + \frac{2}{6}) \approx 0.33$. Same recall, same Precision@6—but less than half the MAP.

WHEN TO USE MAP. It is the workhorse of IR benchmarking (TREC, BEIR): it rewards placing relevant documents early, across the whole list. But MAP shares a fatal limitation with precision and recall: **binary relevance**. Qasim’s perfect-MAP day might have yielded four copper coins where four diamonds were possible. A metric that treats a masterpiece and a merely-acceptable document as identical throws away information.

Metric 5: Area Under the Precision-Recall Curve (AUC-PR). **RETURN TO THE BEACH.** As the child scoops item after item, plot recall on the x -axis (how much of the sand has been searched) and precision on the y -axis (how pure the bucket is). The result is the **Precision–Recall curve**, a downward arc from high-precision/low-recall to low-precision/high-recall. The **area under it** compresses the whole walk into one number: a perfect child scores 1.0; a random one scores the base rate of relevance in the corpus.

Why not the ROC curve? ROC comes from a world of copper wires: voltages down a noisy line, a threshold trading true against false positives. For RAG corpora it is the wrong curve, because relevance is vanishingly rare—six documents in a million. A trivial retriever that returns nothing has a false-positive rate near zero and looks superb on ROC. The PR curve does not have this pathology: a random retriever on a six-in-a-million problem scores $AUC-PR = 6 \times 10^{-6}$, exactly as damning as it should.

WHEN TO USE AUC-PR. Whenever the positive class is rare relative to the corpus—which describes most enterprise RAG, plus fraud detection, security events, and recommenders. It rewards a system that maintains precision even as recall grows, and it is robust to growth in the negative class as your corpus expands.

The cave deepens. Every metric so far treats relevance as binary. But our EVAL set has *graded* scores ($r \in \{0, 1, 2, 3, 4\}$). Return to the cave—now the jewels have differential prices:⁴ finding a diamond at rank 1 is far more valuable than finding silver there, and both beat a diamond buried at rank 20—because the deeper Ali Baba goes, the higher the risk. *The same diamond is worth less the deeper it is found.* This double

⁴ Cave price list: Rock (0), Silver (1), Gold (2), Diamond (4).

sensitivity—to *what* you find and *where* you find it—is exactly what nDCG captures.

Metric 6: Normalised Discounted Cumulative Gain (nDCG). The gold standard—the first metric to be both *rank-aware* and *relevance-weighted*. Unpack the supercalifragilistic name inside-out, in three nested layers plus a normalisation:

LAYER 1: GAIN. The **gain** of a document at rank i is its relevance score, amplified exponentially:

$$G_i = 2^{\text{rel}_i} - 1$$

A rock ($\text{rel} = 0$) has gain 0; silver (1) gain 1; gold (2) gain 3; a diamond (4) gain 15. A diamond is fifteen times a silver, not four—the exponential bakes in the practitioner’s intuition that top-tier relevance matters disproportionately.

LAYER 2: CUMULATIVE GAIN. Sum the gains up to rank K : $\text{CG}_K = \sum_{i=1}^K G_i$. This is the total value of Ali Baba’s knapsack after K shelves.

LAYER 3: DISCOUNTED CUMULATIVE GAIN. Discount each gain by a logarithmic penalty that grows with rank:

$$\text{DCG}_K = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}$$

The deeper the shelf, the higher the risk—so the same gold found at rank 5 is worth less than at rank 1. The logarithm is gentle at the top and harsh at the bottom, modelling the declining patience of a user scanning results.

NORMALISATION. Raw DCG is not comparable across queries—a query with one relevant document can never reach the DCG of one with fifty. Divide by the **Ideal DCG (IDCG)**: the DCG of the same documents sorted by relevance, descending:

$$\text{nDCG}_K = \frac{\text{DCG}_K}{\text{IDCG}_K}$$

Every query lands in $[0, 1]$; perfect ranking scores 1.0; scores average cleanly across the EVAL set.

A WORKED EXAMPLE.⁵ With relevance vector $[1, 3, 0, 2, 1]$:

Two conventions coexist. Järvelin–Kekäläinen (2002) used linear gain $G_i = \text{rel}_i$; Burges (2005) introduced $2^{\text{rel}_i} - 1$, now the default in `scikit-learn` and `pytrec.eval`. When reading an older paper, check the formula—the numbers do not compare across conventions. And if your ground truth supplies continuous relevance in $[0, 1]$, rescale to an integer range first, or use the linear gain; $2^{0.9} \approx 1.87$ vs. $2^{0.6} \approx 1.52$ are barely separable.

A jocular way to explain logarithmic discount: I am willing to promise anyone a million dollars on their 200th birthday. The present value of that promise is not much. The diamond at rank 20 is the same promise; $\log_2$ writes the discount schedule.

⁵ Relevance grades $\{1, 3, 0, 2, 1\}$ at ranks 1–5. Work through the arithmetic once to build muscle memory.

$$\begin{aligned} \text{DCG}_5 &= \frac{2^1 - 1}{\log_2 2} + \frac{2^3 - 1}{\log_2 3} + \frac{2^0 - 1}{\log_2 4} + \frac{2^2 - 1}{\log_2 5} + \frac{2^1 - 1}{\log_2 6} \\ &= 1 + 4.415 + 0 + 1.292 + 0.387 \approx 7.095 \end{aligned}$$

The ideal ordering sorts the grades to [3, 2, 1, 1, 0]:

$$\begin{aligned} \text{IDCG}_5 &= \frac{7}{\log_2 2} + \frac{3}{\log_2 3} + \frac{1}{\log_2 4} + \frac{1}{\log_2 5} + \frac{0}{\log_2 6} \\ &= 7 + 1.893 + 0.5 + 0.431 + 0 \approx 9.824 \end{aligned}$$

So $\text{nDCG}_5 = 7.095/9.824 \approx 0.722$: the system achieved 72% of the best possible ranking of these very documents.

Deep Dive: A second example to try your-

self: relevance vector [0, 1, 0, 2, 1]. You should find $\text{DCG}_5 = 0.631 + 1.292 + 0.387 \approx 2.310$, ideal ordering [2, 1, 1, 0, 0] giving $\text{IDCG}_5 = 3 + 0.631 + 0.5 \approx 4.131$, and $\text{nDCG}_5 \approx 0.56$. The gold at rank 4 is the tragedy: moved to rank 1 it would contribute 3.0 instead of 1.29. One transposition is the difference—and moving it is precisely the reranker's job.

This diagnostic pair is the minimum for any serious RAG team. Both can be computed automatically at every build; the delta between builds is where continuous improvement lives.

⁶Precision/Recall → MRR → MAP → AUC-PR → nDCG. Each metric fixes a limitation of the previous one. Students who understand this progression understand *why* nDCG is the gold standard, rather than blindly accepting it.

WHEN TO USE nDCG. Always, as your primary retrieval metric; the other five are diagnostics. The canonical pairing is nDCG@K alongside Recall@K:

- nDCG low, Recall high: the documents are found but ranked poorly—fix the *reranker*.
- nDCG low, Recall low: the documents are not found at all—fix retrieval: chunking, embedding model, hybrid mix.
- nDCG high, Recall high: celebrate, and move to Act III.

THE PROGRESSION MATTERS.⁶ The six metrics form a staircase. Recall and Precision are binary and position-blind. MRR adds rank-awareness but sees only the first hit. MAP extends rank-awareness to all relevant documents but stays binary. AUC-PR makes the tradeoff continuous and robust to imbalance. nDCG adds graded relevance to rank-awareness—the summit.

Summary: Retrieval Metrics

Binary, not rank-aware: Recall@K, Precision@K, F_1 . Rank-aware, binary relevance: MRR, MAP, AUC-PR. Rank-aware *and* relevance-weighted: nDCG. Use nDCG as your primary metric; the others as diagnostics.

Interlude: Signal from Noise—Benchmarks, Significance, Diversity

BETWEEN THE RETRIEVAL ACT AND THE GENERATION ACT, THREE DISCIPLINES keep the numbers honest: knowing what the public benchmarks are for, knowing when a difference is real, and knowing what the six metrics do not measure at all.

The Benchmark Landscape: BEIR, MTEB, MS MARCO

THREE PUBLIC BENCHMARKS DOMINATE. **BEIR**⁷ aggregates 18 datasets across 9 task categories, with nDCG@10 as its headline metric; its great surprise was that BM25, the classical sparse retriever, still beats dense neural retrievers on many out-of-domain tasks—the very reason the enterprise default is *hybrid* retrieval. **MTEB**⁸ spans 58 datasets across eight task families; its retrieval sub-score is the most-cited number when choosing an embedding model. **MS MARCO**⁹ is the classical passage-ranking benchmark behind the annual TREC Deep Learning tracks.

⁷ Nandan Thakur et al. (2021). “BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models”. In: **NeurIPS Datasets and Benchmarks**

⁸ Niklas Muennighoff et al. (2022). “MTEB: Massive Text Embedding Benchmark”. In: *arXiv preprint arXiv:2210.07316*

⁹ Payal Bajaj et al. (2016). **MS MARCO: A Human Generated Machine Reading Comprehension Dataset**. Technical report. arXiv:1611.09268. Microsoft

Statistical Significance: Is the Improvement Real?

A RITUAL THAT TEAMS ROUTINELY SKIP is checking whether a metric improvement is significant. “Our new reranker improves nDCG by 1.2 points on our eval set.” On 200 queries, that may be noise; on 10,000, 0.3 points may be real. Without a test, you cannot tell—and you will crown the wrong system.

THE PAIRED BOOTSTRAP is the IR community’s default¹⁰: for systems A and B on the same N queries, compute per-query differences $d^{(i)} = s_A^{(i)} - s_B^{(i)}$; resample the differences with replacement 10,000 times; report the 2.5th and 97.5th percentiles as a 95% confidence interval. If the interval excludes zero, the difference is significant. No distributional assumptions; works on any metric. The **permutation test** is the distribution-free alternative: swap $A \leftrightarrow B$ labels at random and ask how often chance beats the observed gap.

WHY? Your deployment gate should include both: pass a domain gold-dataset threshold, *and* place respectably on BEIR or MTEB. The first tests whether the retriever works for your users; the second tests that it is not inexplicably broken.

¹⁰ Mark D. Smucker, James Allan, and Ben Carterette (2007). “A Comparison of Statistical Significance Tests for Information Retrieval Evaluation”. In: **Proceedings of the 16th ACM Conference on Information and Knowledge Management (CIKM)**, pages 623–632

POWER ANALYSIS CLOSES THE LOOP WITH ACT I. To detect a 1-point absolute nDCG improvement with 80% power at $\alpha = 0.05$ typically

The most dangerous confusion is between *statistical* and *practical* significance. With a large enough eval set, trivial improvements become statistically significant; the practical question is whether the improvement is worth the engineering cost. For nDCG, an absolute improvement of 0.02–0.05 is the typical practical threshold in enterprise settings.

¹¹ Jaime Carbonell and Jade Goldstein (1998). “The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries”. In: **Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval**, pages 335–336. DOI: [10.1145/290941.291025](https://doi.org/10.1145/290941.291025)

¹² Charles L. A. Clarke et al. (2008). “Novelty and Diversity in Information Retrieval Evaluation”. In: **Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval**, pages 659–666. For RAG specifically, diversity-aware retrieval tends to improve multi-hop answer quality by 6–12% over vanilla top- K , because the generator gets a broader evidence base. But excessive diversity ($\lambda < 0.3$) hurts simple factoid queries. Tune λ on your gold set.

¹³ Shahul Es et al. (2023). “RAGAs: Automated Evaluation of Retrieval Augmented Generation”. In: *arXiv preprint arXiv:2309.15217*

requires $N \approx 500$ –1,000 queries. The rule of thumb “shoot for 1,000” matches the statistical-power requirement almost exactly.

Retrieval Diversity: The Missing Dimension

OUR SIX METRICS MEASURE RELEVANCE; NONE MEASURES DIVERSITY. A system returning ten near-duplicates of one document can score perfectly on precision, recall, and nDCG, yet serve a poor experience: homogeneous evidence, wasted context, an illusion of consensus that is really redundancy. Maximal Marginal Relevance¹¹ is the canonical fix—iteratively pick the candidate maximising relevance to the query minus similarity to what is already selected:

$$\text{MMR} = \arg \max_{d \in R \setminus S} \left[\lambda \cdot \text{Sim}_1(d, q) - (1 - \lambda) \max_{d' \in S} \text{Sim}_2(d, d') \right]$$

$\lambda = 1$ is ordinary top- K ; production defaults are $\lambda \in [0.5, 0.8]$. For diversity-aware *evaluation*, α -nDCG¹² extends nDCG so repeated coverage of the same subtopic is discounted.

Act III: Beyond Retrieval—Measuring Generation and Faithfulness

THE SIX METRICS ABOVE EVALUATE ONLY THE RETRIEVER. They measure whether the right evidence reached the LLM—the pantry, not the meal. A system can retrieve perfectly relevant documents and still hallucinate; a system can retrieve mediocre documents and still produce a faithful, useful response by reasoning carefully. A master chef with ordinary ingredients can out-cook a clumsy chef with perfect ones. This act crosses the boundary and evaluates the generator.

The RAGAS Framework: A Starting Point

RAGAS (RETRIEVAL-AUGMENTED GENERATION ASSESSMENT) IS THE MOST WIDELY ADOPTED OPEN-SOURCE FRAMEWORK FOR EVALUATING RAG GENERATION QUALITY.¹³ It defines four component metrics:

Faithfulness. Is the generated answer grounded in the retrieved context? RAGAS decomposes the answer into statements, then checks

whether each can be inferred from the context:

$$\text{Faithfulness} = \frac{|\{\text{supported statements}\}|}{|\{\text{all statements}\}|}$$

This is the generation-side analogue of last week’s response grounding. The difference is in the verification mechanism: our conscience used NLI entailment models; RAGAS supports both LLM-based and classical NLI backends. Classical NLI is cheaper and more conservative; LLM entailment is more flexible with paraphrase but prone to the circularity critiqued below.

Answer Relevancy. Does the answer address the question? RAGAS generates n synthetic questions from the answer (“what question would this text answer?”), then measures cosine similarity between those synthetic questions and the original.

Context Precision. Of the chunks passed to the LLM, are the relevant ones ranked higher? A retrieval metric rephrased in the generation frame: did the retriever send the right context *in the right order*?

Context Recall. Does the retrieved context contain all the information needed? RAGAS decomposes the ground-truth answer into claims and checks whether each can be attributed to the retrieved context.

The Five Limitations of RAGAS

PRODUCTION TEAMS WHO DEPLOY RAGAS QUICKLY DISCOVER ITS BOUNDARIES.

Circular LLM dependency. RAGAS uses an LLM to evaluate the output of an LLM. The evaluator shares the biases and blind spots of the generator; a plausible hallucination can pass as faithful. This is the fox guarding the henhouse. Research confirms *self-enhancement bias*: models rate their own outputs higher than equivalent outputs from other models.¹⁴

Context window blindness. RAGAS evaluates the context *actually passed*, not what *could have been* retrieved. If your retriever sent 3 chunks but missed 10 better ones, RAGAS has no visibility. The retrieval metrics of Act II must carry that load.

Our two-gates architecture—bipartite assertion–evidence graph, strictness spectrum, active gap resolution—is considerably more aggressive than RAGAS’s default decomposition, so our pipelines tend to score well on faithfulness. Measure with an external yardstick anyway.

The *Psych* test. In the TV show, Shawn runs for mayor; his friend Gus explains that politicians answer the question they *wish* they had been asked, and the journalist’s job is to reverse the answer into its question and compare. Asked about homelessness, Shawn answers about brown jelly beans; reverse it—“which colour of jelly beans do you like?”—and the cosine distance to the asked question is enormous. Answer relevancy is exactly this construction.

WHY? RAGAS is *necessary*: a RAG team not computing at least these four numbers is not fulfilling the dharma of an AI engineer. But RAGAS is not *sufficient*. The next section explains why.

¹⁴ Peiyi Wang et al. (2024). “Large Language Models Are Not Fair Evaluators”. In: *arXiv preprint arXiv:2305.17926*
An intuition: ask a Ptolemaic astronomer to judge a Ptolemaic prediction—he agrees, for he already holds the theory. For an honest evaluation you need Kepler, not Ptolemy: a different model family as judge.

Evaluator pinning. Always pin the judge to a dated checkpoint; never use a floating alias in production evaluation. Maintain a frozen calibration set of 50–200 human-verified examples; re-run the evaluator periodically; if agreement drifts by more than 0.05 Cohen’s κ , the evaluator has moved under you.

No claim-level granularity. RAGAS statements are coarse. “Einstein was born in Ulm, Germany in 1879 and won the Nobel Prize in 1921” is one statement holding three-plus independently-falsifiable claims; with most clauses supported, the whole statement can pass while a wrong claim rides through inside it.

Score instability across evaluator models. GPT-4, Claude, and Llama produce different RAGAS scores—and vendors silently update models mid-experiment.

No attribution verification. RAGAS checks consistency with the context, not whether the system *cites* the correct passage per claim. Attribution is a separate dimension—one our assertion–evidence graph invested in heavily, and one RAGAS cannot even see.

The RAGAS Verdict

Use RAGAS as a *diagnostic baseline*—the floor below which no RAG system should operate—not as the sole arbiter of quality. Augment it with claim-level evaluation, human spot-checks, and attribution metrics.

Claim-Level Entailment: FActScore and ALCE

THE MOST IMPORTANT ADVANCE IN GENERATION EVALUATION IS THE MOVE FROM ANSWER-LEVEL TO CLAIM-LEVEL. FActScore (EMNLP 2023) operationalises it:

1. **Decompose** the answer into *atomic facts*—claims that cannot be decomposed further. The Einstein sentence becomes: (a) born in Ulm, (b) born in Germany, (c) born 1879, (d) Nobel Prize in physics, 1921.
2. **Verify** each atomic fact against the retrieved documents, via NLI, LLM-as-judge, or exact-match heuristics.
3. **Score** the supported fraction:

$$\text{FActScore} = \frac{|\{\text{supported atomic facts}\}|}{|\{\text{all atomic facts}\}|}$$

¹⁵ Sewon Min et al. (2023). “FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100

WHY CLAIM-LEVEL MATTERS FOR RAG: answers synthesise from multiple passages, so faithfulness becomes per-claim—which claim came from

which passage, and was the attribution correct? The **ALCE benchmark**¹⁶ extends this to citation quality: citation accuracy runs 40–75% without explicit attribution training, 65–70% typical for strong models. Roughly one citation in three is wrong or missing, even when the answer is right. For enterprise RAG, citation accuracy is a deployment gate.

LLM-as-a-Judge 2.0: Specialised Evaluator Models

EVALUATION IS A SPECIFIC TASK—and by every criterion we use for task-specific modelling, an archetypal case for a *specialised* judge rather than a rented frontier model. The early practice—“ask GPT-4 to rate this 1–5”—has matured into a research programme.

Prometheus-2. The open-source evaluator (EMNLP 2024): *direct assessment* (absolute scoring on a user-supplied rubric, accepted at inference time) and *pairwise ranking* (which of two responses is better). Agreement with human judgement: 72–85% across benchmarks including HHH Alignment. Critically, it runs locally—no API calls, no data leakage, no cost per evaluation.¹⁷

Auto-J, JudgeLM, G-Eval. Auto-J shuffles response order as an explicit position-bias defence and scores multiple criteria with structured feedback. JudgeLM’s innovation is its training data: curated human preferences teaching what good *evaluation* looks like. G-Eval uses chain-of-thought and takes a probability-weighted sum over score tokens rather than a greedy argmax, yielding better-calibrated scores.

THE JUDGE BIASES ARE REAL, AND MUST BE UNDERSTOOD.

Position bias: judges prefer whichever response appears first (or last)—documented swings of 40–60 percentage points, with judge-model choice mattering more than task complexity.¹⁸ Mitigation: evaluate both orderings and average.

Length bias: longer responses win regardless of quality; a 500-word mediocrity outscores a 200-word gem. Mitigation: conciseness as an explicit rubric dimension; length-normalised scores.

Self-enhancement bias: models rate their own outputs higher; the effect is large enough to flip rankings.¹⁹ Mitigation: judge from a different model family, or an open evaluator like Prometheus-2.

Style-over-substance: eloquence conflated with correctness. Mitigation: anchor the rubric with examples; require reasoning before scoring.

Tested on biographies generated by ChatGPT, FActScore reported only 58% factual precision. Nearly half the atomic claims in a fluent, confident biography were unsupported.¹⁵

¹⁶ Tianyu Gao et al. (2023). “ALCE: Automatic LLMs’ Citation Evaluation”. In: *arXiv preprint arXiv:2305.14761*

Deep Dive: Claim-level entailment connects

directly to last week’s response grounding. There we decomposed responses into assertions and verified each against evidence with an NLI model. FActScore is the same principle applied more rigorously—finer decomposition, systematic verification, a quantitative score. Grounding is a special case of claim-level evaluation; FActScore is the numerical summary of the assertion–evidence graph.

¹⁷ Seungone Kim et al. (2024). “Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models”. In: *arXiv preprint arXiv:2405.01535*

HHH: honest, helpful, harmless. The right precedence is the physician’s: *harmless first, then honest, then helpful*. An answer can be perfectly honest and harmless and still useless—“ $G_{\mu\nu} = 8\pi T_{\mu\nu}$ ” is no help to a ten-year-old asking what relativity is.

¹⁸ Zhuoyan Li et al. (2025). “Position Bias in Large Language Model Judgments: A Head-to-Tail Analysis”. In: **Proceedings of the International Joint Conference on Natural Language Processing**

¹⁹ Peiyi Wang et al. (2024). “Large Language Models Are Not Fair Evaluators”. In: *arXiv preprint arXiv:2305.17926*

CALIBRATE THE JUDGE AGAINST HUMANS, OR DO NOT DEPLOY IT. Cohen's κ for categorical agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is observed agreement and p_e chance agreement. Landis–Koch thresholds: 0.41–0.60 moderate, 0.61–0.80 substantial. For enterprise RAG, reject any judge configuration with $\kappa < 0.6$. Kendall's τ covers ranking agreement ($\tau \geq 0.5$ good); Krippendorff's α covers multi-rater ensembles.

²⁰ Haitao Li et al. (2024). "LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods". In: *arXiv preprint arXiv:2412.05579*

LLM-as-Judge: The Rule of Three

- (1) Use a different model family for judging than for generating.
- (2) Average over both orderings in pairwise comparisons.
- (3) Never trust a single judge--cross-validate with a second, and spot-check with humans.

Chain-of-thought judging—requiring the reasoning trace before the score—improves human agreement by 8–12 points, at 3–5× token cost.²⁰ The practical pattern: cheap single-token judges for continuous monitoring, chain-of-thought judges for periodic deep evaluation. In daily life we wear slippers at home and save the fancy shoes for going out. Same with judges.

The Four RGB Abilities: What RAGAS Does Not Measure

THE RGB BENCHMARK (AAAI 2024) IDENTIFIED FOUR FUNDAMENTAL ABILITIES THAT RAG GENERATORS MUST POSSESS—ABILITIES RAGAS DOES NOT EVALUATE.²²

Noise Robustness. Can the model ignore retrieved documents that are topically related but do not contain the answer? Ask for the Maxwell–Boltzmann distribution and retrieval surfaces near-misses discussing probability theory and thermodynamics in general. Current LLMs frequently confuse near-miss information with actual answers. Well-built systems exceed 80%; naive systems fall below 50%.

Negative Rejection. When *all* retrieved documents are irrelevant, does the model say "I don't know," or hallucinate? This is the abstention problem—the heart of the epistemic humility we built last week. RGB's experiments: most LLMs answer confidently on nothing. The single most dangerous failure mode in enterprise RAG.

Information Integration. Can the model synthesise across multiple documents to answer a question no single document answers alone? The generation-side analogue of the multi-hop retrieval we met in Week 5's graph traversals.

Counterfactual Robustness. Can the model detect factual errors in retrieved documents? A document says your switch has 480 ports; everything else says 48. Does it flag the discrepancy or blindly follow the context? There is no clean answer—sometimes the document is right and the model stale; sometimes the reverse.

Act IV: The Frontier—Where Evaluation Goes Next

ACTS II AND III GAVE US THE CLASSICAL TOOLS. But the field is moving beyond these foundations, and several frontiers are opening, each driven by a real-world need the classical metrics cannot address. Some are mature enough for production adoption; others are research directions that will shape the next generation of RAG systems.

Frontier 1: Knowing When to Say “I Don’t Know”

A SYSTEM THAT NEVER SAYS “I DON’T KNOW” IS DANGEROUS. An enterprise RAG should behave like a trusted friend: “Here are three things I know, but they are not sufficient; here is what’s missing; let us work it out together.” RAG is a conversation; partial answers with acknowledged gaps invite the user to contribute.

THREE METRICS CAPTURE THIS DIMENSION:

Negative Rejection Rate: present queries the knowledge base cannot answer and measure honest abstention:

$$\text{NRR} = \frac{|\{\text{correctly-abstained queries}\}|}{|\{\text{unanswerable queries}\}|}$$

State of the field: below 30%—systems hallucinate on 70%+ of the queries they should decline. Regulated-industry deployment gates now demand > 70%.

Noise Robustness Rate: the quantitative counterpart of the RGB ability—one relevant document buried among 4–5 topical distractors; does the system extract the right answer?

Abstention Calibration: when the system states a confidence, is it calibrated? Does 80% stated confidence correspond to 80% empirical accuracy? Partition confidences into M bins; ECE²³ is the population-

²¹ Haitao Li et al. (2024). “LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods”. In: *arXiv preprint arXiv:2412.05579*

The cost dimension: evaluating 10,000 queries with a frontier judge costs \$500–\$2,000 per run; a fine-tuned open judge on your own GPU amortises to near-zero. And the survey finding: a 7B judge fine-tuned on 5,000 domain annotations beats prompted GPT-4 in agreement for that domain.²¹

²² Jiawei Chen et al. (2024). “Benchmarking Large Language Models in Retrieval-Augmented Generation”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Volume 38

The rule: *contradictions are first-class citizens*. Surface the conflict, cite both sources, let the user or a review process resolve: “The current specification lists 52 ports (revision B, March 2026), though earlier documentation stated 48.” Last week’s grounding architecture is designed to make exactly this easy.

Deep Dive: To evaluate the four abilities, construct targeted test sets, one ability each: (1) noise robustness: bury the answering document among 3–5 distractors; (2) negative rejection: provide only irrelevant documents and check abstention; (3) information integration: require facts from at least two documents; (4) counterfactual robustness: inject factual errors and check detection. Isolation makes diagnosis precise.

Max Planck’s professor in Munich, Philipp von Jolly, told the young Planck in 1874 that physics was essentially over—nothing left to discover. Planck ignored the advice and founded quantum mechanics. People say the same about AI today. The frontier of RAG evaluation is wide open; you are welcome to write in it.

I have yet to see an enterprise RAG system that honestly says “I don’t know.” Hundreds of students, hundreds of systems—every one, when pressed, admits their system has never emitted those words. It is the mark of a mature RAG, and very few have grown up. Try the Calabi–Yau test: ask a switch-maker’s RAG about Calabi–Yau manifolds, and watch it deliver an enthusiastic lecture from parametric memory that the response gate should have refused.

²³Chuan Guo et al. (2017). “On Calibration of Modern Neural Networks”. In: *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330

Deep Dive: A worked example. 100 answers

in two bins. Bin A: 60 answers, average stated confidence 0.90, empirical accuracy 0.75. Bin B: 40 answers, confidence 0.60, accuracy 0.65. Then $ECE = 0.6 \times |0.75 - 0.90| + 0.4 \times |0.65 - 0.60| = 0.09 + 0.02 = 0.11$. Read the structure: the high-confidence bin is over-confident by 15 points and dominates—the typical LLM signature, and the dangerous zone, because high-confidence answers are the ones users obey.

The visual companion is the *reliability diagram*: bin confidence on the x -axis, bin accuracy on the y -axis; perfect calibration puts every bar on the diagonal. Related measures: MCE (the worst bin’s gap) and the Brier score (a proper scoring rule penalising both miscalibration and lack of sharpness). Standard fixes: temperature scaling (divide logits by a learned T ; preserves accuracy, reshapes confidence), Platt scaling, isotonic regression.

weighted gap between accuracy and confidence per bin:

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

Perfect calibration gives 0; modern LLMs run 0.05–0.15, and larger instruction-tuned models are often *worse*, because instruction tuning pushes confidence up without pushing accuracy up.

Frontier 2: Multi-Hop and Synthesis Evaluation

MANY REAL-WORLD QUESTIONS REQUIRE REASONING ACROSS MULTIPLE DOCUMENTS. “What is the relationship between Company A’s revenue decline and Industry B’s regulatory changes?” Classical metrics evaluate each retrieval independently; they cannot measure *integration*. Three dimensions:

Information integration: construct questions requiring facts from at least two documents; check that both facts appear and are correctly related.

Contradiction detection: inject disagreeing documents; does the answer acknowledge the conflict or silently resolve it (often incorrectly)? The rule from last week stands: contradictions are first-class citizens.

Bridging inference: Document A: drug X inhibits enzyme Y. Document B: enzyme Y is overexpressed in disease Z. The bridge: X may be therapeutic for Z. Construct questions requiring exactly one bridging step.

Frontier 3: Automated Evolutionary Test Set Generation

BUILDING EVAL SETS MANUALLY IS EXPENSIVE, SLOW, AND DOES NOT SCALE. An SME summit produces 100 queries a day; you need 1,000–10,000 for statistical power, covering edge cases human annotators might not imagine.

ARES.²⁷ Automates the pipeline: (1) **synthetic data generation**—from a few seed (query, answer) pairs and in-domain passages, an LLM generates hundreds of examples; (2) **judge fine-tuning**—lightweight classifiers for context relevance, faithfulness, answer relevance; (3) **evaluation with PPI**—prediction-powered inference combines judge predictions with only ~150 human annotations into confidence intervals per metric. Synthetic data handles coverage; human annotation handles calibration.

Giskard RAGET, DeepEval. RAGET generates test sets directly from the knowledge base, at multiple difficulty levels, targeting specific failure modes. DeepEval provides a superset of RAGAS with G-Eval, BLEU/ROUGE, and custom rubric judges.

THE EVOLUTIONARY ANGLE: start from seed questions and apply LLM-driven mutation to generate harder variants:

Add constraints — “What was Q3 revenue?” becomes “What was Q3 revenue for the Enterprise segment in EMEA, excluding one-time charges?”

Require multi-hop — “Who is the CEO?” becomes “Who is the CEO of the company that acquired our largest supplier?”

Inject distractors — add near-miss documents to the context.

Flip expected answers — “Is Product X available in Japan?” where the knowledge base says *no*. Tests negative rejection.

Increase ambiguity — “How many users do we have?” (active? total? paying?) Tests disambiguation.

Frontier 4: Agentic and Trajectory Evaluation

WHEN YOUR RAG SYSTEM IS AN AGENT—PLANNING, RETRIEVING, CALLING TOOLS, ITERATING—EVALUATING ONLY THE FINAL OUTPUT IS DANGEROUSLY INCOMPLETE. Two agents can produce the same correct answer via radically different trajectories: one through careful reasoning, the other through lucky guessing. The lucky guesser fails the next hard question.

TRACE²⁸ proposes a hierarchical trajectory utility: **process efficiency** (steps, tokens, tool calls), **cognitive quality** (planning, hypothesis formation— show me your work), and **evidence grounding** (at each step, retrieved evidence or parametric memory? For enterprise RAG, parametric reliance is a bug). Plus *scaffolded capability assessment*: the minimum guidance needed to succeed, rather than binary pass/fail.

THE THREE-LAYER PRODUCTION FRAMEWORK:

Layer 1: System efficiency—latency, tokens, tool-call count, cost per query; operational, silent on quality. *Layer 2: Session outcomes*—task success,

WHY? ECE operationalises the thesis of the whole session. Negative rejection asks: does it abstain when it should? ECE asks: when it answers, is its stated confidence honest? A system scoring 95% on faithfulness and 0.30 on ECE is right most of the time and overconfident precisely when wrong. In a courtroom, a hospital, a trading floor, that overconfidence is the only metric that matters.

²⁴ Yixuan Tang and Yang Yi (2024). “MultiHop-RAG: Benchmarking Retrieval-Augmented Generation for Multi-Hop Queries”. In: *arXiv preprint arXiv:2401.15391*

²⁵ Hao Li et al. (2025). “HopRAG: Multi-Hop Reasoning for Logic-Aware Retrieval-Augmented Generation”. In: **Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics**

²⁶ Yuankai Jiang et al. (2025). “BRIEF: Bridging Retrieval and Inference for Multi-hop Reasoning via Compression”. In: **Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics**

Benchmarks: HotpotQA, MuSiQue, 2Wiki-MultiHopQA; MultiHop-RAG adds ground-truth evidence chains per hop, so you can score the *path*, not just the answer.²⁴ HoP-RAG (ACL 2025) treats passages as graph vertices and retrieval as traversal²⁵—this frontier speaks directly to Week 5’s city of graphs. BRIEF (NAACL 2025) compresses documents into atomic-proposition summaries and asks whether the answer can be reconstructed from them.²⁶

²⁷ Jon Saad-Falcon et al. (2023). “ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems”. In: *arXiv preprint arXiv:2311.09476*

correctness, satisfaction; the classical end-to-end view. *Layer 3: Node-level precision*—at each decision point: right tool? well-formed query? sound reasoning step? This is where diagnostic power lives.

Frontier 5: Goodhart’s Law and the Danger of Metrics

HERE IS THE SINGLE MOST IMPORTANT CAVEAT OF THE DAY. The moment you post a metric on the wall, people optimise for it. Goodhart (1975), rephrased by Strathern: *when a measure becomes a target, it ceases to be a good measure*. The US No Child Left Behind Act ranked schools on tested reading and math; the documented result was decline in the untested subjects—art, music, history, civics.

RAG-SPECIFIC REWARD-HACKING MODES: trained on faithfulness alone, generators over-hedge (“it is possible that...”—technically faithful, useless); trained on citation recall alone, they over-cite; a length-biased judge breeds over-writing; a judge sharing the generator’s model family colludes. Mitigations: **metric ensembles** (optimise a multi-dimensional bundle), **adversarial evals** (queries designed to break known gaming), **red-team the metric** (humans hunt high-score / low-quality prompts), and **held-out gold judges** never used in training, reserved for drift detection.

Frontier 6: The Cost Dimension and the Conversation

EVALUATION CANNOT ESCAPE ECONOMICS. A \$5, 30-second answer may be operationally worse than an 85%-as-good answer at \$0.05 and 2 seconds. Plot cost-per-query against quality; the **Pareto frontier** is the set of systems no rival beats on both; any system below it should not ship. Four efficiency metrics pair with the quality metrics: cost@quality, quality@cost, tokens-per-correct-answer, and the CFO’s metric—dollars-per-correct-answer.

AND RAG SHIPS AS A CHATBOT, so single-turn metrics miss the conversation: *contextual faithfulness* (consistent with its own earlier turns?), *context-switch handling* (re-retrieve on a topic pivot, or over-anchor?), and *turn efficiency* (how many turns to a satisfactory answer?). CoQA and QuAC are the classical conversational-QA datasets; MT-RAG annotates ground-truth retrievals per turn. User simulation—an LLM plays the user for 5–10 turns—is cheap and high-volume, but watch for simulator hallucination.

Connecting Everything: The Evaluation Map

LET US NOW CONNECT EVERY EVALUATION DIMENSION BACK TO THIS SUMMER’S ARCHITECTURE. Every week built a component. Every component has a failure mode. Every failure mode has a metric:

Embeddings & retrieval (Weeks 1–2) — Retrieval quality is measured directly by Recall@K, Precision@K, MRR, MAP, nDCG.

Reranking (Week 2) — Affects precision and nDCG. A good reranker pushes relevant results up; a bad one leaves them buried.

Chunking (Week 3) — Chunk quality affects context recall. Bad chunks split relevant information across boundaries. Metrics: context recall, information integration.

Derivative artifacts (Week 4) — The understudy’s worth is measured as Recall@K and context precision *before and after* the artifact substitutes for the original.

GraphRAG & multi-hop (Week 5) — Graph reasoning answers to information integration, bridging inference, and the multi-hop benchmarks.

The two gates (Week 6) — The request gate answers to noise robustness; the response gate to negative rejection, faithfulness, and FActScore. The assertion–evidence graph is the mechanism; FActScore is the metric.

The full pipeline (today) — End-to-end metrics (answer correctness, satisfaction, ECE) capture the whole but do not diagnose *where* failures occur. Component-level metrics do.

Essential Readings

FIVE READINGS THAT FORM THE CONCEPTUAL FOUNDATION OF THIS WEEK.

- **Es et al., RAGAs: Automated Evaluation of Retrieval-Augmented Generation (2023)**³⁰

The foundational framework for generation-side RAG evaluation. Required reading to understand the baseline everyone in industry uses—and its limitations.

The evolutionary metaphor is apt: variety through mutation, selection for diagnostic value. The fittest test cases are those that expose the system’s weaknesses; over release cycles the eval set evolves to nip at the heels of production quality, and the yardstick never ossifies.

²⁸ Jingwen Zhou et al. (2026). “TRACE: Trajectory-Aware Comprehensive Evaluation for Deep Research Agents”. In: **Proceedings of the ACM Web Conference 2026**. TRACE showed that outcome-only metrics create an “Illusion of Competence”: agents that score well on final accuracy but reason poorly are fragile in deployment. The trajectory reveals the fragility.

²⁹ Filip Radlinski, Madhu Kurup, and Thorsten Joachims (2008). “How Does Click-through Data Reflect Retrieval Quality?” In: *Proceedings of the 17th ACM Conference on Information and Knowledge Management (CIKM)*, pages 43–52

The working stack: *offline* evaluation (gold dataset + nDCG + RAGAS + FactScore + ECE) gates deployment; *online* evaluation (thumbs-feedback, shadow-mode judges, human spot-checks) drives continuous improvement; interleaving²⁹ compares systems on live traffic with smaller samples than naive A/B. Tooling: LangSmith, Phoenix, Langfuse, W&B Traces—the observability layer our labs run on. The uncomfortable industry number: roughly 70% of RAG systems still lack systematic evaluation.

Kelvin measures; Cameron warns; Goodhart explains *how* the warning comes true. This is also why Act I insisted the eval set be hidden from the development team.

Quality-at-any-cost is the academic metric; quality-on-a-budget is the production one. The semantic cache, the grounding-strictness dial, the reranker choice—evaluate them as *frontier shifts*, not absolute improvements.

Deep Dive: The evaluation map is not

decorative—it is the diagnostic tool that tells you where to invest engineering effort. Faithfulness low but context recall high: the problem is generation, not retrieval. Context recall low: fix chunking or retrieval; no prompt engineering will compensate. Negative rejection low: tighten last week’s gates. nDCG low but FactScore high on the answers that do emerge: the generator copes, but the reranker starves it. ECE high: temperature-scale the confidence. The map turns a single low score into an actionable diagnosis.

³⁰ Shahul Es et al. (2023). “RAGAs: Automated Evaluation of Retrieval Augmented Generation”. In: *arXiv preprint arXiv:2309.15217*

³¹ Sewon Min et al. (2023). “FactScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100

³² Jiawei Chen et al. (2024). “Benchmarking Large Language Models in Retrieval-Augmented Generation”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Volume 38

- Min et al., *FactScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation (EMNLP 2023)*³¹

The paper that established claim-level evaluation. The 58% ChatGPT result is a wake-up call for anyone deploying generative AI without evaluation.

- Chen et al., *Benchmarking Large Language Models in Retrieval-Augmented Generation (AAAI 2024)*³²

The RGB benchmark: noise robustness, negative rejection, information integration, counterfactual robustness—the diagnostic frame for generator evaluation.

- Guo et al., *On Calibration of Modern Neural Networks (ICML 2017)*³³

The canonical reference for ECE, reliability diagrams, and temperature scaling—the machinery beneath abstention calibration.

- Kim et al., *Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models (EMNLP 2024)*³⁴

The best open-source evaluator available: custom rubrics, direct assessment vs. pairwise ranking, evaluation without API dependency.

Oliver Twist’s List of Readings

OPTIONAL, BUT EACH ONE DEEPENS A SPECIFIC FRONTIER.

- Saad-Falcon et al., *ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems (NAACL 2024)*³⁵

The path from 150 human annotations to a full evaluation suite, via synthetic generation and prediction-powered inference.

- Thakur et al., *BEIR (NeurIPS 2021 Datasets Track)*³⁶ and Muenighoff et al., *MTEB (2022)*³⁷

The two public benchmarks worth knowing well—and the BM25 surprise that justifies hybrid retrieval.

- Smucker, Allan, Carterette, *A Comparison of Statistical Significance Tests for Information Retrieval Evaluation (CIKM 2007)*³⁸

Why the paired bootstrap is the IR default; read before you next announce an improvement.

- Li et al., *LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods* (2024)³⁹

The definitive survey of the judge paradigm: biases, mitigations, fine-tuned judges, open problems.

- Tang & Yang, *MultiHop-RAG* (2024)⁴⁰ and Jiang et al., *BRIEF* (NAACL 2025)⁴¹

Multi-hop evaluation with ground-truth evidence chains, and the compression approach to synthesis testing.

- TRACE: *Trajectory-Aware Comprehensive Evaluation for Deep Research Agents* (ACM Web Conference 2026)⁴²

The state of the art in agentic evaluation. Read if you are building or evaluating agent-based RAG.

³³ Chuan Guo et al. (2017). “On Calibration of Modern Neural Networks”. In: *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330

³⁴ Seungone Kim et al. (2024). “Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models”. In: *arXiv preprint arXiv:2405.01535*

³⁵ Jon Saad-Falcon et al. (2023). “ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems”. In: *arXiv preprint arXiv:2311.09476*

A Calibration Note

THIS SESSION WAS DENSE. We built the yardstick—the gold dataset, cherry-picked and graded, versioned and hidden—and named whose job it is: yours. We climbed the staircase from precision and recall through MRR, MAP, and AUC-PR to nDCG, with Sarah and Meera on the beach and Ali Baba and Qasim in the cave. We learned to tell signal from noise with the bootstrap and power analysis. We crossed into generation with RAGAS and its five limitations, sharpened to atomic claims with FActScore and ALCE, put the judges themselves on trial, and measured the four RGB abilities. And we surveyed the frontier: abstention and ECE, multi-hop synthesis, evolutionary test sets, trajectory evaluation, Goodhart’s law, and the Pareto frontier.

The arc of this session mirrors the arc of the summer so far. We started with foundations—geometry, retrieval—just as this session started with precision and recall. We added structure—chunking, artifacts, graphs—just as this session added rank-awareness and graded gain. We built the gates—just as this session built the metrics that watch them. Last week gave the system a conscience; this week gave the conscience a report card.

BUT THE DEEPEST LESSON IS SIMPLER. Measurement is not the end of the process; it is the beginning. A metric without action is trivia. A metric with action is engineering. The evaluation map we drew today is not a report card—it is a diagnostic tool. When faithfulness drops, look

at grounding. When recall drops, look at chunking. When negative rejection drops, look at the gates. The map tells you where to invest your next engineering hour.

AND THE DEEPEST METRIC OF ALL? It is not Recall@K, not faithfulness, not even FActScore. It is the one we have been building toward for two weeks: *does the system know when it does not know?* Last week we built the architecture of epistemic humility—gates that refuse, grounding that verifies. Today we learned to *measure* that humility: negative rejection rate for abstention, noise robustness for distraction, ECE for honesty about confidence. A system that scores 95% on faithfulness but 20% on negative rejection is a system that sounds brilliant until it encounters the one question it cannot answer—and then it lies. In a courtroom, a hospital, a trading floor, that lie is the only metric that matters.

The measure of a system is not its best performance on a curated benchmark. It is its worst performance on a query no one anticipated. Build your eval sets to find those queries before your users do.

Next week, we put the yardstick to work: measurement in hand, the remaining sessions turn from building components to hardening and composing them into systems you would stake a production deployment on.

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